

3.02 Kinematics				
3.02a	Language of kinematics	a) Understand and be able to use the language of kinematics: position, displacement, distance, distance travelled, velocity, speed, acceleration, equation of motion. <i>Learners should understand the vector nature of displacement, velocity and acceleration and the scalar nature of distance travelled and speed.</i>		MQ1

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3.02b 3.02c	Graphical representation	b) Understand, use and interpret graphs in kinematics for motion in a straight line. c) Be able to interpret displacement-time and velocity-time graphs, and in particular understand and be able to use the facts that the gradient of a displacement-time graph represents the velocity, the gradient of a velocity-time graph represents the acceleration, and the area between the graph and the time axis for a velocity-time graph represents the displacement.		MQ2
3.02d 3.02e	Constant acceleration	d) Understand, use and derive the formulae for constant acceleration for motion in a straight line: $v = u + at$ $s = ut + \frac{1}{2}at^2$ $s = \frac{1}{2}(u + v)t$ $v^2 = u^2 + 2as$ $s = vt - \frac{1}{2}at^2$ <i>Learners may be required to derive the constant acceleration formulae using a variety of techniques:</i> - by integration, e.g. $v = \int a dt \Rightarrow v = u + at$, - by using and interpreting appropriate graphs, e.g. velocity against time, - by substitution of one (given) formula into another (given) formula, e.g. substituting $v = u + at$ into $s = \frac{1}{2}(u + v)t$ to obtain $s = ut + \frac{1}{2}at^2$.	e) Be able to extend the constant acceleration formulae to motion in two dimensions using vectors: $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ $\mathbf{s} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$ $\mathbf{s} = \frac{1}{2}(\mathbf{u} + \mathbf{v})t$ $\mathbf{s} = \mathbf{v}t - \frac{1}{2}\mathbf{a}t^2$ <i>Questions set involving vectors may involve either column vector notation, e.g. $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ or $\mathbf{i}, \mathbf{j}$ notation, e.g. $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$.</i> <i>[The formula $\mathbf{v} \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{u} + 2\mathbf{a} \cdot \mathbf{s}$ is excluded.]</i>	MQ3

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3.02f 3.02g	Non uniform acceleration	f) Be able to use differentiation and integration with respect to time in one dimension to solve simple problems concerning the displacement, velocity and acceleration of a particle: $v = \frac{ds}{dt}$ $a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$ $s = \int v dt \text{ and } v = \int a dt$	g) Be able to extend the application of differentiation and integration to two dimensions using vectors: $\mathbf{x} = f(t)\mathbf{i} + g(t)\mathbf{j}$ $\mathbf{v} = \frac{d\mathbf{x}}{dt} = \dot{\mathbf{x}} = f'(t)\mathbf{i} + g'(t)\mathbf{j}$ $\mathbf{a} = \frac{d\mathbf{v}}{dt} = \dot{\mathbf{v}} = \frac{d^2\mathbf{x}}{dt^2} = f''(t)\mathbf{i} + g''(t)\mathbf{j}$ $\mathbf{x} = \int \mathbf{v} dt \text{ and } \mathbf{v} = \int \mathbf{a} dt$ <i>Questions set may involve either column vector or $\mathbf{i}, \mathbf{j}$ notation.</i>	MQ4

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3.02h 3.02i	Gravity		h) Be able to model motion under gravity in a vertical plane using vectors where $\mathbf{a} = \begin{pmatrix} 0 \\ -g \end{pmatrix}$ or $\mathbf{a} = -g\mathbf{j}$. i) Be able to model the motion of a projectile as a particle moving with constant acceleration and understand the limitation of this model. <i>Includes being able to:</i> <i>1. Use horizontal and vertical equations of motion to solve problems on the motion of projectiles.</i> <i>2. Find the magnitude and direction of the velocity at a given time or position.</i> <i>3. Find the range on a horizontal plane and the greatest height achieved.</i> <i>4. Derive and use the cartesian equation of the trajectory of a projectile.</i> <i>[Projectiles on an inclined plane and problems with resistive forces are excluded.]</i>	MQ5

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